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Bihar Engineering University, Patna

B.Tech 7th Semester Exam., 2025

Code: 105701 ***Time: 03 Hours***

Semester: VII ***Subject: Data Science*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) A vector is
(i) A single value
(ii) A graph
(iii) A list of numbers
(iv) A file type

(b) Big Data is characterized by
(i) Only small datasets
(ii) Very low speed
(iii) Manual processing
(iv) Large volume, velocity, variety

(c) NumPy is mainly used for
(i) Web development
(ii) Audio editing
(iii) Numerical computing
(iv) Drawing pictures

(d) Bayes Theorem is related to
(i) Probability
(ii) Algebra
(iii) Cooking
(iv) All of the above

(e) KNN stands for
(i) k- Nearest Neighbors
(ii) None of the above
(iii) Known New Numbers
(iv) Key Neural Network

(f) Correlation measures
(i) Difference between two files
(ii) The mass of an object
(iii) Relationship between variables
(iv) The number of rows

(g) Object recognition is commonly used in
(i) Self-driving cars
(ii) Food delivery
(iii) Gardening
(iv) Sewing machines

(h) Which is a type of supervised learning?
(i) Clustering
(ii) Reinforcement feedback
(iii) Classification
(iv) Random guessing

(i) Which of the following is a classification algorithm?
(i) Linear Regression
(ii) K-Means
(iii) Naïve Bayes
(iv) PCA

(j) Object recognition is related to
(i) Identifying objects in images
(ii) Writing poems
(iii) Drawing charts
(iv) Compressing files

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Q.2 Solve both questions : **[7 x 2 = 14]**

(a) Describe the concept of Data Science, its importance, and real-world applications.

(b) Explain in detail the characteristics of Big Data and challenges in handling it.

Q.3 Solve both questions : **[7 x 2 = 14]**

(a) Discuss the full pipeline of data munging, manipulation, rescaling, and dimensionality reduction.

(b) Explain Python toolkits for Data Science such as NumPy, Matplotlib, and NLTK with examples.

Q.4 Solve both questions : **[7 x 2 = 14]**

(a) Explain vectors and matrices with examples. How are they used in Data Science?

(b) Discuss probability concepts: independence, conditional probability, random variables, and distribution.

Q.5 Solve both questions : **[7 x 2 = 14]**

(a) Explain linear regression in detail, including assumptions, cost function, and regularization methods.

(b) Describe the different types of machine learning with real-world examples.

Q.6 Solve both questions : **[7 x 2 = 14]**

(a) Compare classification and regression with suitable examples.

(b) Discuss any two classification algorithms (e.g., Naive Bayes, KNN, SVM) in detail.

Q.7 Solve both questions : **[7 x 2 = 14]**

(a) Explain stock market prediction with ML, covering regression, time series, and neural network.

(b) Discuss in detail the role of data science in weather forecasting, including data sources and models used.

Q.8 Solve both questions : **[7 x 2 = 14]**

(a) What is the difference between data analysis and reporting, and why is this distinction important in data-driven decision making?

(b) What is data visualization, and what are the purposes and differences of bar charts, line charts, and scatterplots?

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Real-Time Sentiment Analysis

(b) Variety, Velocity, and Veracity in Big Data

(c) Reinforcement Learning

(d) Object Recognition Systems

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